

Workshop 4: solutions

1. Neither model is a GLM. In a generalized linear model, the link function relates the mean of the response, $\mu_i = E(y_i)$, to the linear predictor $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$. In both models given in the question, the linear predictor is equated to the random variable itself, or to a transformation of it, rather than to its mean. Hence the models cannot be written in the standard GLM form. Moreover, this specification would imply that the response (or its transformation) is completely determined by the predictors, leaving no random variation, which is incompatible with the assumed Gamma distribution.
2. Since the response is a count, a natural choice is the Poisson distribution, $y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i)$, for $i = 1, \dots, n$. The expected value is given by

$$\mu_i \equiv \mathbb{E}(y_i) = a^2 + 2bax_i + b^2x_i^2 = (a + bx_i)^2.$$

To obtain a linear predictor in the parameters, we use the square-root link function

$$g(\mu_i) = \sqrt{\mu_i}.$$

The GLM is therefore

$$\sqrt{\mu_i} = a + bx_i,$$

with linear predictor $\eta_i = a + bx_i$, which in matrix form is given by

$$\boldsymbol{\eta} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}.$$

Note that since $\sqrt{\mu_i} \geq 0$, this model implicitly requires $a + bx_i \geq 0$.

3. Taking logarithms gives

$$\log(\mu_i) = \log(n) + \log(\gamma_k) + \log(\alpha_j), \text{ if } y_i \text{ is gender } k \text{ and belief } j.$$

Define

$$\tau = \log(n), \quad \beta_k = \log(\gamma_k), \quad \delta_j = \log(\alpha_j).$$

Then,

$$\log(\mu_i) = \tau + \beta_k + \delta_j.$$

Assuming $y_i \stackrel{\text{ind}}{\sim} \text{Poi}(\mu_i)$, this corresponds to a GLM with log link, since $g(\mu_i) = \log(\mu_i) = \eta_i$. If the data are ordered FB, FN, MB, MN (Female Believer, Female Non-believer, Male Believer, Male Non-believer), the linear predictor is

$$\begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tau \\ \beta_1 \\ \beta_2 \\ \delta_1 \\ \delta_2 \end{pmatrix}.$$

This model is not identifiable since the sum of the second and third columns of the design matrix equals the first column, and similarly the sum of the fourth and fifth columns equals the first column. Setting $\beta_1 = \delta_1 = 0$ (or $\beta_2 = \delta_2 = 0$) gives the identifiable model

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \tau \\ \beta_2 \\ \delta_2 \end{pmatrix}.$$

4. A ‘logistic regression’ model might be appropriate (as used for the heart attack data in the notes). Treating the six O-rings at a given launch as independent binomial trials, each with probability p_i of failure at temperature temp_i , we can write

$$y_i \stackrel{\text{ind.}}{\sim} \text{binom}(6, p_i),$$

where y_i denotes the number of damaged O-rings in the i th launch. Using the logit link function gives

$$\text{logit}(p_i) = \log \left(\frac{p_i}{1 - p_i} \right) = \beta_1 + \beta_2 \text{temp}_i.$$

The corresponding model matrix therefore has i th row given by $(1, \text{temp}_i)$.

5. We begin by writing $f(y; \lambda)$ as $\exp\{\log f(y; \lambda)\}$:

$$\log f(y; \lambda) = \log \lambda - \lambda y \Rightarrow \exp\{\log f(y; \lambda)\} = \exp\{\log \lambda - \lambda y\} = \exp\left\{\frac{y(-\lambda) - (-\log \lambda)}{1} + 0\right\}.$$

Hence the distribution belongs to the exponential family with

- $\theta = -\lambda \Rightarrow \lambda = -\theta$. Note that because $\lambda > 0$, we have that $\theta < 0$.
- $b(\theta) = -\log \lambda = -\log(-\theta)$.
- $a(\phi) = \phi = 1$.
- $c(y, \phi) = 0$.

We have established that $\mu \equiv E(Y) = b'(\theta)$, and in this case, $b'(\theta) = -\frac{1}{\theta}$.

Sanity check: The expectation of an exponentially distributed variable is known to be $\mu \equiv E(Y) = 1/\lambda$. We have derived that $E(Y) = -1/\theta$ and that $\theta = -\lambda$. Consequently, $E(Y) = 1/\lambda$, confirming consistency. Note that, in this case, the distribution was not parameterized in terms of its mean. Furthermore, we have that

$$b''(\theta) = \frac{1}{\theta^2} \Rightarrow \text{var}(Y) = b''(\theta) \times a(\phi) = \frac{1}{\theta^2} \times 1 = \frac{1}{\theta^2} = \frac{1}{\lambda^2} = \mu^2.$$

Note that $V(\mu) = \mu^2$. This is the variance function of the Gamma family; indeed, the exponential distribution is a special case of the Gamma distribution (with shape parameter equal to 1).

6. (a) By definition,

$$\mu = \mathbb{E}(Y) = \sum_{y=0}^1 y f(y; p) = 0 \times (1 - p) + 1 \times p = p.$$

(b) Since $\mathbb{E}(Y) = \mu = p$, we can write the probability mass function as

$$f(y; \mu) = \mu^y (1 - \mu)^{1-y}, \quad y \in \{0, 1\}, \quad 0 < \mu < 1.$$

Taking logarithms,

$$\log f(y; \mu) = y \log \mu + (1 - y) \log(1 - \mu).$$

Rearranging gives

$$\begin{aligned} \exp\{\log f(y; \mu)\} &= \exp\{y \log \mu + (1 - y) \log(1 - \mu)\} \\ &= \exp\{y \log \mu + \log(1 - \mu) - y \log(1 - \mu)\} \\ &= \exp\left\{y \log\left(\frac{\mu}{1 - \mu}\right) + \log(1 - \mu)\right\} \\ &= \exp\left\{\frac{y \log\left(\frac{\mu}{1 - \mu}\right) - [-\log(1 - \mu)]}{1} + 0\right\}. \end{aligned}$$

This is the exponential family form with:

- $\theta = \log\left(\frac{\mu}{1 - \mu}\right) \Rightarrow \mu = \frac{e^\theta}{1 + e^\theta}$.
- $b(\theta) = -\log(1 - \mu) = \log(1 + e^\theta)$.
- $a(\phi) = \phi = 1$.
- $c(y, \phi) = 0$.

7. We have that

$$\begin{aligned} \exp\{\log f(y; \pi, n)\} &= \exp\left\{\log\binom{n}{ny} + ny \log \pi + (n - ny) \log(1 - \pi)\right\} \\ &= \exp\left\{\log\binom{n}{ny} + ny \log\left(\frac{\pi}{1 - \pi}\right) + n \log(1 - \pi)\right\} \\ &= \exp\left\{\frac{y \log\left(\frac{\pi}{1 - \pi}\right) - (-\log(1 - \pi))}{1/n} + \log\binom{n}{ny}\right\}, \end{aligned}$$

which has the form of a density/pmf of a distribution in the exponential family with:

- $\theta = \log\left(\frac{\pi}{1 - \pi}\right)$, which means that $\pi = \frac{e^\theta}{1 + e^\theta}$.
- $b(\theta) = -\log(1 - \pi) = \log(1 + e^\theta)$.
- $a(\phi) = 1/n$. Note that here $\phi = 1$ and $\omega = n$.
- $c(y, \phi) = \log\binom{n}{ny}$.

8. (a) As before, let us write $f(y; \mu, \gamma) = \exp\{\log f(y; \mu, \gamma)\}$. We have that

$$\begin{aligned} \log f(y; \mu, \gamma) &= \frac{1}{2} \log(\gamma) - \frac{1}{2} \log(2\pi y^3) - \frac{\gamma(y - \mu)^2}{2\mu^2 y} \\ &= \frac{1}{2} \log(\gamma) - \frac{1}{2} \log(2\pi y^3) - \frac{\gamma(y^2 - 2\mu y + \mu^2)}{2\mu^2 y} \\ &= \frac{1}{2} \log(\gamma) - \frac{1}{2} \log(2\pi y^3) - \frac{\gamma y}{2\mu^2} + \frac{\gamma}{\mu} - \frac{\gamma}{2y} \end{aligned}$$

and so

$$\begin{aligned}\exp\{\log f(y; \mu, \gamma)\} &= \exp\left\{\frac{1}{2}\log(\gamma) - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma y}{2\mu^2} + \frac{\gamma}{\mu} - \frac{\gamma}{2y}\right\} \\ &= \exp\left\{\frac{y[-1/(2\mu^2)] - (-1/\mu)}{1/\gamma} + \left(\frac{1}{2}\log \gamma - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma}{2y}\right)\right\}\end{aligned}$$

This is in the exponential family of distributions with:

- $\theta = -\frac{1}{2\mu^2}$ (note that $\theta < 0$) $\Rightarrow \mu = \sqrt{-\frac{1}{2\theta}}$.
- $b(\theta) = -\frac{1}{\mu} = -\sqrt{-2\theta}$.
- $a(\phi) = \phi = \frac{1}{\gamma}$.
- $c(y, \phi) = \frac{1}{2}\log \gamma - \frac{1}{2}\log(2\pi y^3) - \frac{\gamma}{2y} = -\frac{1}{2}\log(\gamma^{-1}2\pi y^3) - \frac{\gamma}{2y} = -\frac{1}{2}\left[\log(\phi 2\pi y^3) + \frac{1}{\phi y}\right]$.

(b) From the properties of the exponential family we have that

$$\mathbb{E}(Y) = b'(\theta), \quad \text{and} \quad \text{var}(Y) = b''(\theta)a(\phi).$$

From part (a) we have that $b(\theta) = -\sqrt{-2\theta}$ and so

$$b'(\theta) = \frac{d}{d\theta}(-\sqrt{-2\theta}).$$

I will do the derivative step by step to avoid any mistakes. Letting $y = -\sqrt{v}$ and $v = -2\theta$, we get

$$\frac{d}{d\theta}(-\sqrt{-2\theta}) = \frac{dy}{d\theta} = \frac{dy}{dv} \frac{dv}{d\theta} = \left(\frac{d}{dv}(-\sqrt{v})\right) \left(\frac{d}{d\theta}(-2\theta)\right) = \left(-\frac{1}{2}v^{-1/2}\right)(-2) = v^{-1/2}.$$

Because $v = -2\theta$ we have

$$v^{-1/2} = (-2\theta)^{-1/2} = \frac{1}{\sqrt{-2\theta}} = \sqrt{-\frac{1}{2\theta}} = \mu.$$

So we conclude that $\mathbb{E}(Y) = \mu$. This should not come as a surprise, as it was already stated in the question text that μ is the mean.

We just need now to obtain the second derivative of $b(\theta)$. Letting $y = \frac{1}{\sqrt{v}} = v^{-1/2}$ and $v = -2\theta$ we have

$$b''(\theta) = \frac{d}{d\theta}b'(\theta) = \frac{dy}{d\theta} = \frac{dy}{dv} \frac{dv}{d\theta} = \left(\frac{d}{dv}v^{-1/2}\right) \left(\frac{d}{d\theta}(-2\theta)\right) = \left(-\frac{1}{2}v^{-3/2}\right)(-2) = v^{-3/2}.$$

Therefore,

$$v^{-3/2} = (-2\theta)^{-3/2} = \left(\sqrt{-\frac{1}{2\theta}}\right)^3 = \mu^3.$$

Finally, and recalling that in part (a) we have found that $a(\phi) = 1/\gamma$, we have

$$\text{var}(Y) = b''(\theta)a(\phi) = \frac{\mu^3}{\gamma}.$$

(c) The canonical link function is such that $g(\mu) = \theta$. Therefore, the canonical link function is

$$g(\mu) = -\frac{1}{2\mu^2}.$$

9. We have that

$$\exp\{\log f(y; \nu)\} = \exp\left\{\log \Gamma\left(\frac{\nu+1}{2}\right) - \frac{1}{2}\log(\nu\pi) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{\nu+1}{2}\log\left(1 + \frac{y^2}{\nu}\right)\right\}$$

As we can see, the term $\frac{\nu+1}{2}\log\left(1 + \frac{y^2}{\nu}\right)$ is the only term involving both the parameter and the data and cannot be factored into a product of a function of y and a function of ν . Hence the t -distribution does not belong to the exponential family.

10. Note that $Y_i \in \{0, 1, 2, 3, \dots\}$ and, as such,

$$Z_i = \begin{cases} 1, & \text{if } Y_i > 0, \\ 0, & \text{if } Y_i = 0, \end{cases}$$

and hence it holds that

$$\begin{aligned} \pi_i &= \Pr(Z_i = 1) \\ &= 1 - \Pr(Z_i = 0) \\ &= 1 - \Pr(Y_i = 0). \end{aligned}$$

Because $Y_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\mu_i)$, we know that

$$\Pr(Y_i = 0) = \frac{e^{-\mu_i} \mu_i^0}{0!} = e^{-\mu_i},$$

and hence

$$\pi_i = 1 - e^{-\mu_i}.$$

Solving for μ_i we get

$$1 - \pi_i = e^{-\mu_i} \Rightarrow \mu_i = -\log(1 - \pi_i).$$

Because $\log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$, we have that

$$\log(\mu_i) = \log(-\log(1 - \pi_i)) = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

Note that $\mathbb{E}(Z_i) = 1 \times \Pr(Z_i = 1) + 0 \times \Pr(Z_i = 0) = \Pr(Z_i = 1) = \pi_i$. This link function is known as the complementary log-log link (covered in more detail in Workshop 5).

11. We first compute the expectation of z_i :

$$\begin{aligned} \mathbb{E}(z_i) &= \mathbb{E}[g'(\mu_i)(y_i - \mu_i) + \mathbf{x}_i^\top \boldsymbol{\beta}] \\ &= g'(\mu_i)[\mathbb{E}(y_i) - \mu_i] + \mathbf{x}_i^\top \boldsymbol{\beta} \\ &= g'(\mu_i)(\mu_i - \mu_i) + \mathbf{x}_i^\top \boldsymbol{\beta} \\ &= \mathbf{x}_i^\top \boldsymbol{\beta}, \end{aligned}$$

where we have used the fact that $g'(\mu_i)$ and $\mathbf{x}_i^\top \beta$ are constants and that $\mathbb{E}(y_i) = \mu_i$. $\mathbb{E}(z_i) = \mathbf{x}_i^\top \beta$ and hence $\mathbb{E}(\mathbf{z}) = \mathbf{X}\beta$. Next we compute the variance of z_i :

$$\begin{aligned}\text{var}(z_i) &= \text{var}(g'(\mu_i)(y_i - \mu_i) + \mathbf{x}_i^\top \beta) \\ &= \text{var}(g'(\mu_i)y_i - g'(\mu_i)\mu_i + \mathbf{x}_i^\top \beta) \\ &= g'(\mu_i)^2 \text{var}(y_i) \\ &= g'(\mu_i)^2 V(\mu_i) \phi.\end{aligned}$$

Because the y_i are independent and each z_i depends only on y_i , the z_i are also independent. Therefore the covariance matrix of $\mathbf{z}$ is diagonal with entries $g'(\mu_i)^2 V(\mu_i) \phi$. Using the notation from the lecture notes (Section 11.2), $g'(\mu_i)^2 V(\mu_i) = w_i^{-1}$, where $w_i = 1$ as implicit in the text of the question, so the covariance matrix of $\mathbf{z}$ can be written as $\mathbf{W}^{-1} \phi$ where $\mathbf{W}$ is a diagonal matrix, with $W_{ii} = w_i$.

12. (a)

$$\frac{d}{d\beta} \sum_{i=1}^n w_i (z_i - \beta x_i)^2 = -2 \sum_i w_i (z_i - x_i \beta) x_i$$

Setting this to zero and re-arranging yields:

$$\hat{\beta} = \frac{\sum_i w_i z_i x_i}{\sum_i w_i x_i^2}.$$

Differentiating the weighted sum of squares again shows $\hat{\beta}$ minimizes it.

(b) The link function is the identity function, $g(\mu_i) = \mu_i$, and so $g'(\mu_i) = 1$. For the Poisson distribution we have $V(\mu_i) = \mu_i$ (see Section 11.1 of the notes). Hence, at each IRLS iteration we will have

$$w_i = (g'(\mu_i)^2 V(\mu_i))^{-1} = \frac{1}{\mu_i},$$

and

$$z_i = g'(\mu_i)(y_i - \mu_i) + x_i \beta = (y_i - \mu_i) + x_i \beta = (y_i - x_i \beta) + x_i \beta = y_i.$$

Note that $g(\mu_i) = \eta_i = x_i \beta$, but because g is the identity function, $g(\mu_i) = \mu_i = \eta_i = x_i \beta$.

So the IRLS steps will proceed as follows.

- **Initialize** Set $\mu^{(0)} = (1, 1, 3, 7, 4)^\top$ (i.e. set $\mu^{(0)}$ to $\mathbf{y}$, except when $y_i = 0$, since μ_i needs to be greater than zero). $\eta^{(0)} = \mu^{(0)}$, since the model has an identity link.
- **Iteration 1** First find the iterative weights and working/pseudo responses:

$$\begin{aligned}w_i^{(0)} &= (\mu_i^{(0)})^{-1} \Rightarrow (w_1^{(0)}, w_2^{(0)}, w_3^{(0)}, w_4^{(0)}, w_5^{(0)}) = \left(1, 1, \frac{1}{3}, \frac{1}{7}, \frac{1}{4}\right), \\ z_i^{(0)} &= y_i \Rightarrow (z_1^{(0)}, z_2^{(0)}, z_3^{(0)}, z_4^{(0)}, z_5^{(0)}) = (0, 1, 3, 7, 4).\end{aligned}$$

Using part (a) we have that $\beta^{(1)} = \frac{\sum_{i=1}^5 w_i^{(0)} z_i^{(0)} x_i}{\sum_i w_i^{(0)} x_i^2}$. I will do this calculation in R.

```
omega_0 <- c(1, 1, 1/3, 1/7, 1/4)
z_0 <- c(0, 1, 3, 7, 4)
x <- c(1, 2, 3, 4, 5)
beta_1 <- sum(omega_0*z_0*x) / sum(omega_0*(x^2))
beta_1
## [1] 0.8466523
```

- **Iteration 2** We have that $w_i^{(1)} = (\mu_i^{(1)})^{-1} = (x_i \beta^{(1)})^{-1}$ and $z_i^{(1)} = y_i$. We can then calculate $\beta^{(2)}$ as in the first iteration.

```
mu_1 <- x*beta_1
omega_1 <- 1/mu_1
z_1 <- c(0, 1, 3, 7, 4)
beta_2 <- sum(omega_1*z_1*x) / sum(omega_1*(x^2))
beta_2
## [1] 1
```

Actually, $\hat{\beta} = 1$ is the MLE — in this case the IRLS has converged in 2 steps. Let us confirm this.

```
x <- c(1, 2, 3, 4, 5)
y <- c(0, 1, 3, 7, 4)
res_toy <- glm(y ~ x-1, family = poisson(link = "identity"))
res_toy$coefficients
## x
## 1
```

- (c) From general theory, we know that (asymptotic distribution of the maximum likelihood estimate $\hat{\beta}$) the covariance matrix for a GLM with scale parameter (ϕ) equal to 1 is $(\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}$. For the current single parameter model, this becomes $(\sum_i w_i x_i^2)^{-1}$, with the square root of this giving the standard error for $\hat{\beta}$. To evaluate it we need the final estimates for the w_i :

$$\begin{array}{l|ccccc} \hat{\mu}_i = \hat{\eta}_i = x_i \hat{\beta} = x_i & 1 & 2 & 3 & 4 & 5 \\ w_i = \hat{\mu}_i^{-1} & 1 & 1/2 & 1/3 & 1/4 & 1/5 \end{array}$$

which results in a standard error estimate for $\hat{\beta}$ of 0.258. We can check this is indeed the correct value by comparing to the value returned by the `glm` function.

```
sqrt(vcov(res_toy))
## x
## x 0.2581989
```